Lecture Exam Statistics AI 2020S

tags: `Exam` `Statistics` `2020S`

Example 1a-c)

- a) Explain with a few words, the difference between an ordinal attribute and a quantitative attribute in descriptive statistics. State one example for each kind of these attributes. [4 points]
- b) Explain with a few words, what is meant by a "type I error" and a "type II error" in inductive statistics. [4 points]
- c) Explain with a few words, what is meant by the term "confidence level" in the context of confidence intervals in inductive statistics. [4 points]

Example 1d)

Mark **only** the **correct** statements. None, one or more of the 4 given statements may be correct. [4 points, no minus points for wrong answers]

Select one or more:

- [ ] The covariance has a range from 0 to $+\infty$.
- [ ] The standard error is a summary statistic which is completely independent from the sample size.
- [ ] The median is an appropriate measure of location for ordinal data.
- [ ] Especially for nominal data the mean is an appropriate summary statistic because of its efficient use of the available information.

3.

Example 2)

Out of a box containing 5 red and 6 green balls, a sample of 4 balls is drawn with replacement. Calculate the exact probability that there is exactly 1 green ball in the drawn sample. [10 points]

20.49

Example 3a-b)

During a mountain climbing tour of 6 days, the climbed distances for each day (rounded to whole meters) were recorded: 214 ; 179 ; 195 ; 222 ; 24 ; 178

- a) Calculate the average climbed distance and the median climbed distance. [7 points]
- b) Calculate the variance and the standard deviation of this population. [8 points]

- a)
  - mean = 168.667
  - median = 187
- b)
  - varPop = 4452.56
  - stDevPop = 66.7275

Example 4)

Sue and Bob are on a shooting range. They shoot alternately. Sue is the better shooter with a hit rate of 90%. Bob has a hit rate of 70%. Someone hears a shot, but does not see the person shooting. There is no hit on the target. With what probability did Bob shoot, if Sue and Bob are the only possible shooters? [15 points]

(1 - bob's hit rate) / (1 - sue's hit rate + 1 - bob's hit rate)

75%

Example 5a-b)

The television time in hours per day and the weight in kilograms of a population of 15 children were examined. The following results were obtained:

	mean	$\sum_{i=1}^{15}(x_i - \bar{x})^2$	covariance
television time	4.733	42.933	15.973
weight	52.400	2561.600	

Furthermore, it is known that there is a linear relationship between television time and weight.

- a) Using the above information, show that the variance of the television time is 2.862 h^2 and the variance of the weight 170.773 kg^2. (Note: numbers are rounded to 3 decimal places, the 15 children are considered as a population) [8 points]
- b) Determine a linear regression model to predict the weight of a child with 6 hours of television time. State the predicted weight and the parameters of the linear regression model. [12 points]

- a)
  - 1/15*42933 = 2.8622
  - 1/15*2561.600=170.773
- b) y=a+b*x_i
  - b = Cov(x ,y)/Var(x) -> 15.973/2.862 = 5.581
  - a= y_mean - b *x_mean = 52.400 - 5.581 *4.733 = 25.985
  - y = 25.985 + 5.581 *x_i
  - 25.985+5.581*6=59.4715 kg

Example 6a-b)

The effect of a drug on diastolic blood pressure should be investigated. For this purpose, the drug is administered to a sample of 6 patients.

diastolic blood pressure [mmHg]					
71	73	74	82	90	60

- a) Calculate a 95% confidence interval for the "true" mean diastolic blood pressure based on the above sample. The unbiased and consistent point estimator for the mean has already been calculated and is 75 mmHg. [10 points]
- b) Based on the above sample, test the research question that the diastolic blood pressure under the influence of this drug, is statistically significant different from the reference value of 70 mmHg. Formulate the corresponding statistical hypotheses. State the test statistic, the critical value and the test decision. This statistical test should have a type I error of α = 5%. The data can be considered to be normally distribution. [14 points]

- a)
  - sd = 10.198
  - [64.2978 - 85.7022]
- b=
  - Hypot.:
    - H0 μ=70
    - H1 μ != 70
  - Test: T(x)=1.20096
  - t_5,0.957=2.57
  - 1.2 < 2.57 -> null Hupo. cannot be rejected
  - pVal: 0.28355 > α 5% -> cannot be rejected